Varun Gumma

Ph.D. Student · Nanyang Technological University

varun.gumma@protonmail.com | varungumma.netlify.app | Google Scholar
github.com/VarunGumma | x.com/VarunGumma23

EDUCATION

Nanyang Technological University (NTU) — Aug 2025 – Present
Ph.D., School of Electrical and Electronic Engineering — *Singapore*
GPA: 5.0/5 · Advisor: Prof. Soujanya Poria

Indian Institute of Technology (IIT) Madras — Aug 2021 – Jun 2023
M.Tech. in Computer Science & Engineering — *Chennai, India*
GPA: 9.62/10 · Department Rank: 3 · Advisors: Dr. Raj Dabre, Prof. Mitesh Khapra
Teaching Assistant: Pattern Recognition & Machine Learning, Linear Algebra & Random Processes, Fundamentals of Deep Learning

Birla Institute of Technology and Science (BITS) Pilani — Aug 2017 – Jun 2021
B.E. in Computer Science, Minor in Physics — *Hyderabad, India*
GPA: 9.14/10 · Advisors: Prof. N.L. Bhanu Murthy, Prof. Barsha Mitra
Teaching Assistant: Introductory Physics, Computer Programming, Discrete Mathematics, Data Structures & Algorithms, Foundations of Data Science, Machine Learning

EXPERIENCE

Microsoft Research — May 2023 – Jul 2025
SCAI Center Fellow · Advisors: Dr. Kalika Bali, Dr. Sunayana Sitaram — *Bangalore, India*
Worked on multi-dimensional evaluation of Large Language Models and building Machine Translation systems for low-resource languages.

National University of Singapore — Jun 2022 – Jul 2022
Visiting Research Scholar · Advisor: Dr. Aditya Karanam — *Singapore*
Worked on developing Machine Learning and Deep Learning models to extract *suggestions* and identify *suggested features* from noisy user comments using Named Entity Recognition.

Defence Research & Development Organisation (DRDO), Research Centre Imarat — May 2019 – Jul 2019
Student Intern — *Hyderabad, India*
Worked on building an Object Detection model to identify cars in a given image.

PUBLICATIONS

C = Conference, W = Workshop, J = Journal, A = arXiv preprint * equal contribution, † equal mentorship

[A.3] **SemField: A Simple, Linear, Continuous, yet Robust Semantic Watermark**
Varun Gumma, Navonil Majumder, Soujanya Poria
arXiv preprint — **arXiv**

[A.2] **IDEAgent: Agentic Quality-Diversity Search for Research Idea Generation**
Varun Gumma, Navonil Majumder, Soumitra Sinhahajari, Soujanya Poria
arXiv preprint — **arXiv**

[C.11] **UPDESH: Synthesizing Grounded Instruction Tuning Data for 13 Indic Languages**
Pranjal A. Chitale, **Varun Gumma**, Sanchit Ahuja, Prashant Kodali, Manan Uppadhyay, Deepthi Sudharsan, Sunayana Sitaram
64th Annual Meeting of the Association for Computational Linguistics — **ACL '26**

[C.10] **OffTopicEval: When Large Language Models Enter the Wrong Chat, *Almost Always!***
Jingdi Lei*, **Varun Gumma***, Rishabh Bhardwaj*, Seok Min Lim, Chuan Li, Amir Zadeh, Soujanya Poria
14th International Conference on Learning Representations — **ICLR '26**

[C.9] **Towards Inducing Long-Context Abilities in Multilingual Neural Machine Translation Models**
Varun Gumma*, Pranjal A. Chitale*, Kalika Bali
2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics — **NAACL '25**

[W.3] **Beyond Metrics: Evaluating LLMs' Effectiveness in Culturally Nuanced, Low-Resource Real-World Scenarios**
Millicent Ochieng, **Varun Gumma**, Sunayana Sitaram, Jindong Wang, Vishrav Chaudhary, Keshet Ronen, Kalika Bali, Jacki O'Neill
6th Workshop on African Natural Language Processing — **AfricaNLP '25**

[A.1] **HEALTH-PARIKSHA: Assessing RAG Models for Health Chatbots in Real-World Multilingual Settings**
Varun Gumma, Anandita Raghunath, Mohit Jain†, Sunayana Sitaram†
arXiv preprint **arXiv**

[W.2] **Contamination Report for Multilingual Benchmarks**
Sanchit Ahuja*, **Varun Gumma***, Sunayana Sitaram
Evaluating Evaluations: Examining Best Practices for Measuring Broader Impacts of Generative AI **EvalEval '24**

[C.8] **PARIKSHA: A Large-Scale Investigation of Human-LLM Evaluator Agreement on Multilingual and Multi-Cultural Data**
Ishaan Watts, **Varun Gumma**, Aditya Yadavalli, Vivek Seshadri, Manohar Swaminathan, Sunayana Sitaram
2024 Conference on Empirical Methods in Natural Language Processing **EMNLP '24**

[C.7] **Akal Badi ya Bias: An Exploratory Study of Gender Bias in Hindi Language Technology** 🏆
Rishav Hada, Safiya Husain, **Varun Gumma**, Harshita Diddee, Aditya Yadavalli, Agrima Seth, Nidhi Kulkarni, Ujwal Gadiraju, Aditya Vashistha, Vivek Seshadri, Kalika Bali
7th Annual ACM Conference on Fairness, Accountability, and Transparency **FAccT '24**

[C.6] **METAL: Towards Multilingual Meta-Evaluation**
Rishav Hada*, **Varun Gumma***, Mohamed Ahmed, Kalika Bali, Sunayana Sitaram
Findings of the Association for Computational Linguistics: NAACL 2024 **NAACL '24**

[C.5] **MEGAVERSE: Benchmarking Large Language Models Across Languages, Modalities, Models and Tasks**
Sanchit Ahuja, Divyanshu Aggarwal, **Varun Gumma**, Ishaan Watts, Ashutosh Sathe, Millicent Ochieng, Rishav Hada, Prachi Jain, Mohamed Ahmed, Kalika Bali, Sunayana Sitaram
2024 Annual Conference of the North American Chapter of the Association for Computational Linguistics **NAACL '24**

[W.1] **MunTTS: A Text-to-Speech System for Mundari**
Varun Gumma, Rishav Hada, Aditya Yadavalli, Pamir Gogoi, Ishani Mondal, Vivek Seshadri, Kalika Bali
7th Workshop on Computational Methods for Endangered Languages **ComputEL '24**

[C.4] **MAFIA: Multi-Adapter Fused Inclusive LanguAge Models**
Prachi Jain*, Ashutosh Sathe*, **Varun Gumma**, Kabir Ahuja, Sunayana Sitaram
18th Conference of the European Chapter of the Association for Computational Linguistics **EACL '24**

[C.3] **Are Large Language Model-based Evaluators the Solution to Scaling Up Multilingual Evaluation?**
Rishav Hada, **Varun Gumma**, Adrian de Wynter, Harshita Diddee, Mohamed Ahmed, Monojit Choudhury, Kalika Bali, Sunayana Sitaram
Findings of the Association for Computational Linguistics: EACL 2024 **EACL '24**

[J.1] **IndicTrans2: Towards High-Quality and Accessible Machine Translation Models for all 22 Scheduled Indian Languages**
Jay Gala*, Pranjal A. Chitale*, Raghavan AK, **Varun Gumma**, Sumanth Doddapaneni, Aswanth Kumar, Janki Nawale, Anupama Sujatha, Ratish Puduppully, Vivek Raghavan, Pratyush Kumar, Mitesh M. Khapra, Raj Dabre, Anoop Kunchukuttan
Transactions on Machine Learning Research **TMLR '23**

[C.2] **An Empirical Study of Leveraging Knowledge Distillation for Compressing Multilingual Neural Machine Translation Models**
Varun Gumma, Raj Dabre, Pratyush Kumar
24th Annual Conference of the European Association for Machine Translation **EAMT '23**

[C.1] **PAMMELA: Policy Administration Methodology using Machine Learning**
Varun Gumma, Barsha Mitra, Soumyadeep Dey, Pratik Shashikantbhai Patel*, Sourabh Suman*, Saptarshi Das, Jaideep Vaidya
19th International Conference on Security and Cryptography **SECRYPT '22**

AWARDS AND ACHIEVEMENTS

NTU Research Scholarship · To pursue a full-time Ph.D. as an international student 2025
Outstanding Reviewer, EMNLP 2024 · For notable contributions as a peer reviewer 2024
IIT Madras STAR TA · For outstanding contributions as a Graduate Teaching Assistant 2021 – 2023
GATE CS & IT · All India Rank 159, with a score of 816/1000 2021
BITS Merit Scholarship · For meritorious academic performance in 2017 – 2018 2018

ACADEMIC SERVICE

Peer Reviewer ACL Rolling Review, ICLR '25, ICLR '26, ICLR '27